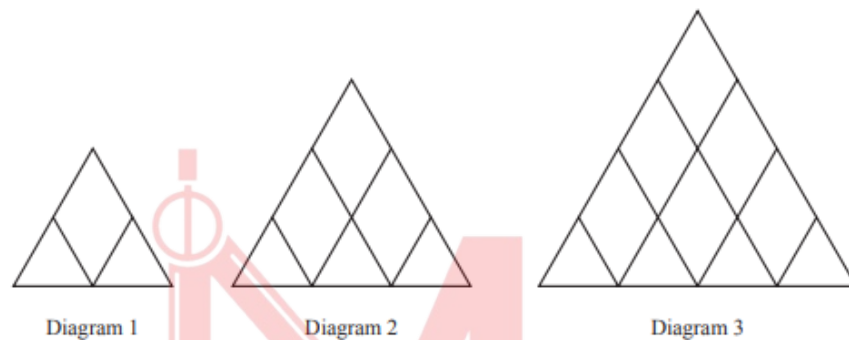


Number Sequences

Worksheet 4 (Calculator)

1. S16/qp43/q10



Each diagram is made from tiles in the shape of equilateral triangles and rhombuses. The length of a side of each tile is 1 unit.

a. Complete the table below for this sequence of diagrams.

Diagram	1	2	3	4	5
Number of equilateral triangle shaped tiles	2	3	4	5	6
Number of rhombus shaped tiles	1	3	6		
Total number of tiles	3	6	10		
Number of 1 unit lengths	8	15	24		

[6]

b.

i. The number of 1 unit lengths in Diagram n is $n^2 + 4n + p$. Find the value of p . [2]

ii. Calculate the number of 1 unit lengths in Diagram 10. [1]

c. The total number of tiles in Diagram n is $an^2 + bn + 1$. Find the value of a and the value of b . [5]

- d. Part of the Louvre museum in Paris is in the shape of a square-based pyramid made from glass tiles. Each of the triangular faces of the pyramid is represented by Diagram 17 in the sequence.
- Calculate the total number of glass tiles on one triangular face of this pyramid. [2]
 - 11 tiles are removed from one of the triangular faces to create an entrance into the pyramid. Calculate the total number of glass tiles used to construct this pyramid. [1]

2. S17/qp41/q9

- The n th term of a sequence is $8n - 3$.
 - Write down the first two terms of this sequence. [1]
 - Show that the number 203 is not in this sequence. [2]
- Find the n th term of these sequences.
 - 13, 19, 25, 31, [2]
 - 4, 8, 14, 22, [2]
- ..., 20, 50, ...

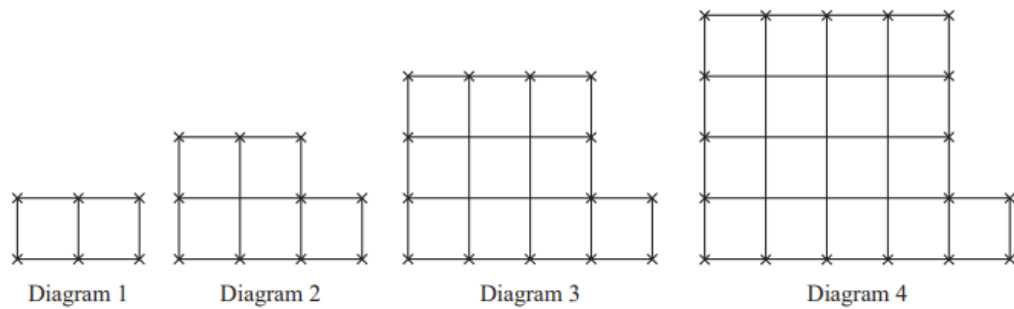
The second term of this sequence is 20 and the third term is 50. The rule for finding the next term in this sequence is subtract y then multiply by 5. Find the value of y and work out the first term of this sequence. [4]

3. S17/qp43/q11

The table shows the first four terms in sequences A, B, C and D. Complete the table. [12]

Sequence	1st term	2nd term	3rd term	4th term	5th term	n th term
<i>A</i>	16	25	36	49		
<i>B</i>	5	8	11	14		
<i>C</i>	11	17	25	35		
<i>D</i>	$\frac{3}{2}$	$\frac{4}{3}$	$\frac{5}{4}$	$\frac{6}{5}$		

4. W17/qp41/q6



These are the first four diagrams in a sequence. Each diagram is made from small squares and crosses.

- a. Complete the table. [6]

Diagram	1	2	3	4	5		n
Number of crosses	6	10	14				
Number of small squares	2	5	10				

- b. Find the number of crosses in Diagram 60. [1]
c. Which diagram has 226 squares? [1]
d. The side of each small square has length 1 cm. The number of lines of length 1 cm in Diagram n is $2n^2 + 2n + q$. Find the value of q . [2]

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5. M18/qp42/q11

The table shows the first five terms of sequences A, B and C.

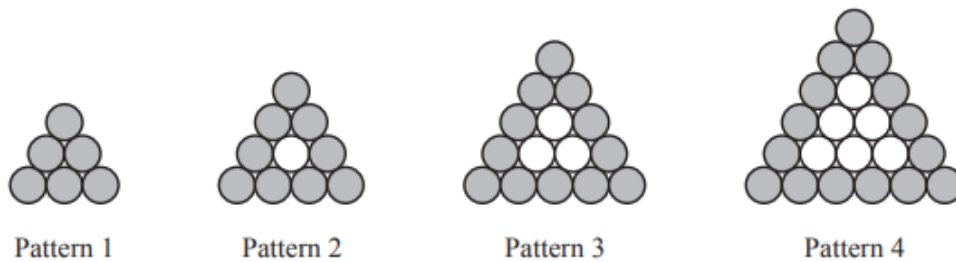
Sequence	1st term	2nd term	3rd term	4th term	5th term	6th term
<i>A</i>	0	1	4	9	16	
<i>B</i>	4	5	6	7	8	
<i>C</i>	-4	-4	-2	2	8	

- a. Complete the table. [3]
- b. Find an expression for the n th term of
 - i. sequence A, [2]
 - ii. sequence B. [1]
- c. Find the value of n when the n th term of sequence A is 576. [2]
- d.
 - i. Find an expression for the n th term of sequence C. Give your answer in its simplest form. [3]
 - ii. Find the value of the 30th term of sequence C. [2]

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6. S18/qp41/q12

Marco is making patterns with grey and white circular mats.



The patterns form a sequence. Marco makes a table to show some information about the patterns.

Pattern number	1	2	3	4	5
Number of grey mats	6	9	12	15	
Total number of mats	6	10	15	21	

- Complete the table for Pattern 5. [2]
- Find an expression, in terms of n , for the number of grey mats in Pattern n . [2]
- Marco makes a pattern with 24 grey mats. Find the total number of mats in this pattern. [2]
- Marco needs a total of 6 mats to make the first pattern. He needs a total of 16 mats to make the first two patterns. He needs a total of $\frac{1}{6}n^3 + an^2 + bn$ mats to make the first n patterns. Find the value of a and the value of b . [6]

7. W18/qp43/q10

- Find the next term and the n th term of this sequence.

$$\frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \frac{6}{11}, \frac{7}{13}$$

- Find the n th term of each sequence.

- $-1, -3, -5, -7, -9,$

- $2, 9, 28, 65, 126,$

8. M19/qp42/q11

a. The table shows the first five terms of sequence A and sequence B.

Term	1	2	3	4	5	6
Sequence A	7	13	23	37	55	
Sequence B	1	3	9	27	81	

i. Complete the table for the 6th term of each sequence. [2]

ii. Find the n th term of

a. sequence A, [2]

b. sequence B. [2]

b. The n th term of another sequence is $4n^2 + n + 3$.

Find

i. the 2nd term, [1]

ii. the value of n when the n th term is 498. [3]

Answer Key

1. S16/qp43/q10

10 (a)	10 15		
	15 21		
	35 48	6	B1 for each correct entry
		2	M1 for any correct substitution in $n^2 + 4n + p$ = number of tiles eg $2^2 + 4(2) + p = 15$
(b) (i)	3		
(ii)	143	1FT	FT $140 + \text{their } (b)(i)$
(c)	$a = \frac{1}{2}$ oe $b = \frac{3}{2}$ oe nfw	5	B1 for a correct simplified equation e.g. $a + b + 1 = 3$, $4a + 2b + 1 = 6$, $9a + 3b + 1 = 10$ etc B1 for a 2 nd correct simplified equation M1 for correctly eliminating one variable for their equations in a and b
			A1 for $a = \frac{1}{2}$ nfw A1 for $b = \frac{3}{2}$ nfw
(d) (i)	171	2FT	FT $\text{their } a \times 17^2 + \text{their } b \times 17 + 1$ M1 for $\text{their } a \times 17^2 + \text{their } b \times 17 + 1$
(ii)	673	1FT	FT $\text{their } (d)(i) \times 4 - 11$

2. S17/qp41/q9

9(a)(i)	5 and 13	1	
9(a)(ii)	$8n - 3 = 203$	M1	Evaluation of 25th or 26th term with supporting evidence or explanation
	25.75 or $25\frac{3}{4}$	A1	Second evaluation of 25th or 26th terms with supporting evidence or explanation If zero scored, SC1 for 25.75 or 197 and 205 with partial evidence or explanation
9(b)(i)	$6n + 7$ oe final answer	2	B1 for $6n + c$ or $kn + 7$ $k \neq 0$
9(b)(ii)	$n^2 + n + 2$ oe final answer	2	B1 for a quadratic expression or second difference = 2
9(c)	$[y =] 10$	2	M1 for $5(20 - y) = 50$
	[First term =] 14	2	M1 for $5(x - \text{their } y) = 20$ or for $20 \div 5 + \text{their } y$

3. S17/qp43/q11

11	64 $(n+3)^2$ oe final answer	1, 2	M1 for a quadratic expression seen or second differences 2
	17 $3n+2$ oe final answer	1, 2	B1 for $3n+k$ (any k) or $kn+2$ ($k \neq 0$)
	47 $(n+3)^2 - (3n+2)$ oe isw	1, 2FT	FT their difference expressions $A - B$ M1 for expression $an^2 + bn + c$ seen or second differences 2
	$\frac{7}{6}$ $\frac{n+2}{n+1}$ oe final answer	1, 2	B1 for $\frac{n+k+1}{n+k}$ seen

4. W17/qp41/q6

6(a)	18 22 $4n+2$ oe 17 26 n^2+1 oe	6	B2 for 18, 22, 17, 26 or B1 for two or three correct values AND B2 for $4n+2$ oe or B1 for $4n+k$ oe or $pn+2$ ($p \neq 0$) AND B2 for n^2+1 oe or B1 for n^2+k oe
6(b)	242	1	FT their $4n+2$ provided a linear expression
6(c)	15	1	
6(d)	3	2	M1 for $2 \times 1^2 + 2 \times 1 + q = 7$ oe

5. M18/qp42/q11

11(a)	25 9 16	3	B1 for each
11(b)(i)	$(n-1)^2$ oe	2	B1 for any quadratic of form $[1]n^2[+bn+c]$
11(b)(ii)	$n+3$ oe	1	
11(c)	25	2	M1 for their $(n-1)^2 = 576$
11(d)(i)	$n^2 - 3n - 2$ final answer	3	M1 for their $(n-1)^2 - their(n+3)$ oe or 2nd diff = 2 soi B1 for $n^2 - n - n + 1$ or better or $-n - 3$ or for expression of form $n^2 - 2n - n + k$ or correct expression not in simplest form
11(d)(ii)	808 cao	2	M1 for substituting 30 in their (d)(i)

6. S18/qp41/q12

12(a)	18 28	2	B1 for each
12(b)	$3n + 3$ oe	2	B1 for $3n + k$ oe or $cn + 3$ oe $c \neq 0$
12(c)	45	2	M1 for identifying 7th pattern or M1 for <i>their</i> $(3n + 3) = 24$
12(d)	$[a =]\frac{3}{2}$ oe $[b =]\frac{13}{3}$ oe	6	M1 for any correct substitution e.g. $\frac{1}{6}(2)^3 + 2^2a + 2b$ A1 for one of e.g. $\frac{1}{6} + a + b = 6$ oe $\frac{8}{6} + 4a + 2b = 16$ oe $\frac{27}{6} + 9a + 3b = 31$ oe $\frac{64}{6} + 16a + 4b = 52$ oe A1 for another of the above M1 for correctly eliminating one variable from <i>their</i> equations A1 for $a = \frac{3}{2}$ A1 for $b = \frac{13}{3}$ oe

7. W18/qp43/q10

10(a)	$\frac{8}{15}$	B1	
	$\frac{n+2}{2n+3}$ oe	B2	B1 for $n+2$ as numerator or $2n+3$ as denominator
10(b)(i)	$1-2n$ oe	2	B1 for $-2n+k$ oe or $pn+1$ ($p \neq 0$) oe
10(b)(ii)	n^3+1 oe	2	M1 for cubic expression

8. M19/qp42/q11

11(a)(i)	77 243	2	B1 for each
11(a)(ii)(a)	$2n^2+5$ oe	2	M1 for a quadratic expression as the answer or B1 for common 2nd difference of 4
11(a)(ii)(b)	3^{n-1} oe	2	B1 for 3^k oe where k is a linear function of n
11(b)(i)	21	1	
11(b)(ii)	11	3	B2 for $(4n+45)(n-11)$ seen or B1 for $4n^2+n+3=498$ oe

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